Robot Mechanics Analysis: 2-DOF Polar Manipulator (R-P)

1. System Description

The robot consists of a **Revolute-Prismatic (RP)** configuration:

- **Joint 1 (Base):** Revolute joint driven by an MG996R servo. Rotates the entire assembly about the Z-axis (Yaw).
- **Joint 2 (Waist):** Prismatic joint driven by a second MG996R servo with a rack-and-pinion mechanism. Translates the end-effector radially.

System Parameters

- $\theta(t)$: Base rotation angle (Generalized Coordinate q_1).
- $r(t)$: Rack extension length (Generalized Coordinate q_2).
- m: Mass of the moving rack assembly (Given: **20g** or **0.02 kg**).
- I_{base}: Moment of inertia of the rotating base housing.
- r_p: Radius of the pinion gear (approx 1.5 cm).

2. Kinematics

2.1 Forward Kinematics (FK)

Given the joint variables (θ, r), the position of the end-effector (x, y) in the Cartesian plane is:

$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$

2.2 Inverse Kinematics (IK)

Given a desired coordinate (x, y), the required joint variables are:

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \text{atan2}(y, x)$$

2.3 Velocity Kinematics (The Jacobian)

The relationship between joint velocities $(\dot{\theta}, \dot{r})$ and tip velocities $(\dot{x}, \dot{y})$ is given by the Jacobian matrix J:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -r\sin(\theta) & \cos(\theta) \\ r\cos(\theta) & \sin(\theta) \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{r} \end{bmatrix}$$

Determinant of Jacobian: $|J| = -r$.

- **Singularity:** The robot loses a degree of freedom when $r = 0$ (The arm is at the exact center).

3. Lagrangian Dynamics (Equation of Motion)

We use the Lagrangian method $L = T - V$ to find the torque/force requirements.

3.1 Kinetic Energy (T)

The kinetic energy consists of the base's rotation and the rack's motion. The rack has velocity components:

- Radial velocity: $v_r = \dot{r}$
- Tangential velocity: $v_t = r\dot{\theta}$

$$T = \frac{1}{2} I_{base} \dot{\theta}^2 + \frac{1}{2} m(\dot{r}^2 + (r\dot{\theta})^2)$$

3.2 Potential Energy (V)

Assuming the robot operates on a horizontal table (Top-Down view), gravity acts perpendicular to the motion:

$$V = 0$$

(Note: If the robot lifts the arm vertically, $V = mgr\sin(\theta)$).

3.3 Lagrangian (L)

$$L = T - V = \frac{1}{2} I_{base} \dot{\theta}^2 + \frac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2)$$

3.4 Euler-Lagrange Equations

Equation: $\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_i}) - \frac{\partial L}{\partial q_i} = \tau_i$

Joint 1: Base Torque (τ_θ)

1. $\frac{\partial L}{\partial \dot{\theta}} = (I_{base} + mr^2)\dot{\theta}$
2. $\frac{d}{dt}(\dots) = (I_{base} + mr^2)\ddot{\theta} + 2mr\dot{r}\dot{\theta}$
3. $\frac{\partial L}{\partial \theta} = 0$

Equation of Motion (Base):

$$\tau_\theta = (I_{base} + mr^2)\ddot{\theta} + \underbrace{2mr\dot{r}\dot{\theta}}_{\text{Coriolis Torque}}$$

- **Insight:** The term $2mr\dot{r}\dot{\theta}$ is the **Coriolis Effect**. If you extend the rack ($\dot{r} > 0$) while spinning ($\dot{\theta} > 0$), the motor feels an extra "drag" trying to slow the rotation down.

Joint 2: Rack Force (F_{rack})

1. $\frac{\partial L}{\partial \dot{r}} = m\dot{r}$
2. $\frac{d}{dt}(\dots) = m\ddot{r}$
3. $\frac{\partial L}{\partial r} = mr\dot{\theta}^2$

Equation of Motion (Rack):

$$F_{rack} = m\ddot{r} - \underbrace{mr\dot{\theta}^2}_{\text{Centrifugal Force}}$$

- **Insight:** The term $mr\dot{\theta}^2$ is the **Centrifugal Force**. As the robot spins faster, the rack naturally "wants" to fly outward. The servo must fight this force to pull the rack in.

4. Hardware Calculation (MG996R)

Motor Specifications

- **Model:** MG996R Digital Servo
- **Stall Torque (τ_{stall}):** ~10 kg-cm = **0.98 N·m** (at 6V)
- **Operating Speed:** 0.17 sec/60° $\approx$ 60 RPM $\approx$ **6 rad/s**

Rack and Pinion Calculation

- **Pinion Radius (r_p):** 1.5 cm = 0.015 m
- **Rack Mass (m):** 20g = 0.02 kg

Force Capability of Rack Joint

The linear force $F_{available}$ generated by the servo on the rack:

$$F_{available} = \frac{\tau_{stall}}{r_p} = \frac{0.98 \text{ N} \cdot \text{m}}{0.015 \text{ m}} \approx 65.3 \text{ Newtons}$$

Load Analysis

Let's assume an extreme case:

- Spinning at max speed: $\dot{\theta} = 6 \text{ rad/s}$

- Full extension: $r = 15 \text{ cm} = 0.15 \text{ m}$

Force Required (Centrifugal):

$$F_{centrifugal} = mr\dot{\theta}^2 = 0.02 \times 0.15 \times 6^2 = 0.02 \times 0.15 \times 36 = \mathbf{0.108} \text{ N}$$

Conclusion

Safety Factor:

$$\text{Safety Factor} = \frac{\text{Available Force}}{\text{Required Force}} = \frac{65.3}{0.108} \approx 600$$

Result: The MG996R is **extremely overpowered** for a 20g rack. It will handle the load easily, even at